

A. BABAEI

PhD Applicant in Neuroscience and Neural Engineering

Biomedical Engineer | Experimental Neuroscience | Scientific Computing

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ACADEMIC PROFILE

Biomedical engineer with graduate training in bioelectric engineering and research experience spanning preclinical neuroscience, neuromodulation, behavioral analysis, scientific programming, signal and image processing, and machine learning. **Seeking PhD training in neuroscience, neural engineering, neuromodulation, or computational neurotechnology.**

RESEARCH INTERESTS

Primary: Neuromodulation; deep brain stimulation; Parkinsonian animal models; behavioral neuroscience; neural engineering

Computational: Scientific Python; MATLAB; repeated-measures analysis; signal and image processing; machine learning; reproducible visualization

Translational: Experimental design, quantitative behavioral phenotyping, scientific instrumentation, and biomedical technology

EDUCATION

M.Sc. Biomedical Engineering (Bioelectric) | Iran University of Science and Technology 2023-2026
Tehran, Iran

- Focus: neural engineering, neuromodulation, machine learning, biomedical signal processing, and image processing.
- Graduate research integrated preclinical experimentation, behavioral assessment, stimulation protocols, and computational analysis.

B.Sc. Biomedical Engineering | Islamic Azad University, Tehran East 2019-2022
Tehran, Iran

- GPA: 18.68/20.
- Contributed to the launch of the university scientific journal and participated in indexed research outputs.

RESEARCH EXPERIENCE

Graduate Researcher - Neuromodulation and Behavioral Neuroscience | Master's Research Project 2023-2026
Tehran, Iran

- Investigated parafascicular thalamic stimulation in a unilateral 6-OHDA rat model of Parkinsonian motor dysfunction.
- Performed or supported stereotaxic procedures, lesion validation, electrode implantation workflows, deep brain stimulation, and behavioral testing.
- Developed quantitative behavioral-analysis workflows using video scoring and customized computational pipelines.
- Applied repeated-measures statistical models, dose-response analysis, data visualization, and reproducibility-oriented reporting in Python.
- Prepared manuscript sections, supplementary tables, scientific figures, graphical abstracts, and submission materials for high-impact journals.

Biomedical Signal, Image, and Machine-Learning Projects | Graduate and Independent Projects

- Worked with MATLAB and Python for preprocessing, feature extraction, classification, visualization, and analysis of biomedical data.
- Developed educational and research code related to neural signals, image processing, behavioral datasets, and mathematical animation.

RESEARCH METHODS AND TECHNICAL SKILLS

Experimental: Rodent handling; stereotaxic workflows; 6-OHDA models; behavioral tests; DBS protocols; experimental documentation

Programming: Python, MATLAB; data cleaning; statistical modeling; automation; scientific figure generation

Analysis: Repeated-measures data; generalized estimating equations; mixed models; model comparison; multiple-testing control

Biomedical computing: Signal processing; image processing; machine learning; video-based behavioral analysis

Scientific communication: Manuscript preparation; systematic data extraction; graphical abstracts; presentation design; research teaching

TEACHING AND ACADEMIC COMMUNICATION

Instructor - Practical MATLAB for Neuroscience | Amirkabir University of Technology, International Neuroscience School

Nov 2025

Tehran, Iran

- Designed and delivered six practical sessions (12 hours) focused on MATLAB applications in neuroscience.

Course Developer and Instructor - MATLAB Beginner to Advanced | Tech Academy

Dec 2023-Present

Tehran, Iran

- Produced a structured 40-hour offline course comprising 38 sessions.
- Developed lesson plans, code examples, presentations, exercises, and neuroscience-oriented applications.

RESEARCH OUTPUTS

- Two ISI-indexed articles reported during the M.Sc. period; full bibliographic entries were not included in the source CV and must be inserted before submission.
- Two ISC-indexed articles reported during the B.Sc. period; full bibliographic entries must be inserted before submission.
- Research code and analytical work are available through GitHub: github.com/A-Babaei

RELEVANT TECHNICAL AND PROFESSIONAL EXPERIENCE

Medical Equipment Supervisor Engineer (Part-Time) | Peyramoon Teb Co.

Dec 2023-Mar 2026

Tehran, Iran

- Conducted device inspections, compliance checks, technical assessment, installation support, maintenance coordination, and troubleshooting.

Medical Equipment Repair Technician | Vistatech-Pishro & Vista Co.

Mar 2023-May 2023

Tehran, Iran

- Completed hands-on work with sterilization systems, autoclaves, ultrasonic devices, dental handpieces, and basic laboratory equipment.

SELECTED DIGITAL AND EDUCATIONAL PROJECTS

Sirdent.ir: Founder and website administrator; product management, technical operations, SEO, medical/dental content, and customer relations

Academy-tech.ir: Technical education platform; website administration, course production, scientific content, and SEO

Scientific portfolio: Figures, graphical abstracts, presentation systems, video editing, websites, and reproducible code